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METHOD OF FABRICATING TRANSISTOR HAVING EXPANDED GATE STRUCTURE, AND TRANSISTOR FABRICATED THEREBY

Abstract

A method of fabricating a semiconductor device includes forming a dummy gate and gate spacers; forming an interlayer dielectric layer; forming a trench having a lower portion and an upper portion that is wider than the lower portion; and forming a gate structure in the trench. The gate structure has an upper portion in the upper portion of the trench, and a lower portion in the lower portion of the trench. The upper portion of the gate structure is wider than the lower portion of the gate structure. A first gate spacer has a first inner sidewall facing the lower portion of the gate structure. A second gate spacer has a second inner sidewall facing the lower portion of the gate structure. The first inner sidewall is apart from the second inner sidewall by a first distance. The upper portion of the gate structure is wider than the first distance.

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Background/Summary

BACKGROUND

[0001] Dimensions of transistors have decreased as process nodes have advanced and integrated circuits have become more dense. The down-scaling of transistors to smaller dimensions allows more transistors to be integrated into a given area and is generally advantageous but also presents challenges as some features become very small.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 is a flowchart of operations in a method of fabricating a semiconductor device according to some embodiments.

[0004] FIGS. 2A-2H are cross-sectional views of a transistor structure at various stages in a manufacturing process according to some embodiments.

[0005] FIGS. 3A and 3B are cross-sectional views of the transistor structure of FIG. 2G according to some embodiments.

[0006] FIG. 4 is a cross-sectional view of a transistor structure according to some embodiments.

[0007] FIG. 5 is a cross-sectional view of a transistor structure according to some embodiments.

[0008] FIGS. 6A-6C are cross-sectional views of transistor structures according to some embodiments.

DETAILED DESCRIPTION

[0009] This disclosure describes embodiments and examples of the subject matter set forth herein and, although specific examples of components, materials, values, steps, arrangements, or the like may be described, such examples are not limiting and other components, materials, values, steps, arrangements, or the like are contemplated.

[0010] As used herein, a term preceded by “a” or “an” (and “the” when antecedent basis is “a” or “an”) indicates both singular and plural of such term, unless indicated otherwise.

[0011] Further, like numbers are intended to denote like elements throughout this disclosure and the drawings, but like numbers or other referential descriptors do not imply a specific hierarchy or order. Likewise, references to “first,” “second,” “third,” or the like do not imply a specific order.

[0012] Further, a description of a first element being “on” a second element may include a case in which the first element is directly on the second element, i.e., the first and second elements are in direct contact, and may also include a case in which an additional element is between the first and second elements, e.g., a case in which the first and second elements are not in direct contact.

[0013] Further, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” and variations thereof indicate a non-exclusive inclusion. For example, a process, article, or apparatus that “comprises” a list or set of stated elements is not limited to only the stated elements, and may include other elements not expressly listed or stated.

[0014] Further, the term “or” is inclusive, not exclusive, such that the term “or” means “and/or”

unless indicated otherwise. Thus, “A or B” means “A and/or B” and encompasses A alone, B alone, and both A and B, unless indicated otherwise.

[0015] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures but do not imply a fixed orientation. The spatially relative terms are intended to encompass different orientations in addition to the orientation depicted in the figures.

[0016] Further, “source/drain(s)” may refer to a source or a drain, individually or collectively dependent upon the context.

[0017] In some embodiments, a transistor includes a gate structure adjacent to a channel region; source and drain regions adjacent to the channel region; a gate contact connected to the gate structure; and source/drain contacts connected to the source/drain regions. In some embodiments, the gate structure includes a conductive (i.e., electrically conductive) fill material (i.e., an electrically conductive material) at least partially surrounded by a work function (WF) layer, with the conductive fill material and the work function layer being separated from a channel region by a gate dielectric layer. In some embodiments, insulating (i.e., electrically insulating) gate spacers are disposed on sides of the gate dielectric layer and/or the gate structure. In some embodiments, the gate spacers are surrounded by a dielectric layer. In some embodiments, the dielectric layer includes an interlayer dielectric (ILD) layer. In some embodiments, the gate structure has an insulating layer thereon. In some embodiments the gate structure has an upper portion and a lower portion, and the upper portion of the gate structure is wider than the lower portion of the gate structure and overhangs the gate spacers. In some embodiments, the conductive fill material has an upper portion and a lower portion, and the upper portion of the conductive fill material is wider than the lower portion of the conductive fill material. The design of the transistor helps to enable fine tuning of the profile of the gate structure to reduce gate resistance (R_g) and helps to enable a device which has large pitch but provides high speed in comparison with other approaches.

[0018] FIG. 1 is a flowchart of operations in a method of fabricating a semiconductor device according to some embodiments, and FIGS. 2A-2G are cross-sectional views of a transistor structure at various stages in a manufacturing process according to some embodiments.

[0019] Operations in FIG. 1 are part of a method **100** that forms a transistor using a gate-last process, e.g., a replacement metal gate (RMG) process, according to some embodiments.

[0020] In some embodiments, the transistor includes a field-effect transistor (FET). In some embodiments, the transistor includes a metal-oxide-semiconductor field-effect transistor (MOSFET). In some embodiments, the transistor includes a planar transistor, a finFET transistor, or the like.

[0021] Referring to FIGS. 1 and 2A, an operation **110** includes forming a dummy gate **270** on a fin **215** of a substrate **210**, and forming gate spacers **240** on sidewalls of the dummy gate **270**.

[0022] In some embodiments, the substrate **210** is a semiconductor substrate. In some embodiments, the substrate **210** is a bulk semiconductor substrate, a silicon-on-insulator (SOI) substrate, a semiconductor-on-insulator substrate, or the like. In some embodiments, the substrate **210** is doped with an n-type or p-type dopant, or is undoped. In some embodiments, the substrate **210** is or includes a semiconductor wafer such as a single-crystalline semiconductor wafer that is a section of a single crystal semiconductor ingot. In some embodiments, the substrate **210** includes a buried oxide layer.

[0023] In some embodiments, the fin **215** is a semiconductor material. In some embodiments, the fin **215** is of the same material as an uppermost region of the substrate **210**. In some embodiments, the fin **215** includes silicon. In some embodiments, the transistor is a MOSFET and the dummy gate is formed on the substrate **210** rather than on the fin **215**.

[0024] In some embodiments, the fin **215** is formed by reducing a thickness of the substrate **210** in regions adjacent to the fin **215**. In some embodiments, the fin **215** is formed by etching or

patterning the substrate **210**. In some embodiments, the fin **215** is formed by patterning and etching the substrate **210** using a photolithography process that includes depositing a layer of photoresist material on the substrate, irradiating or exposing the photoresist material in accordance with a pattern corresponding to the fin **215**, developing the photoresist material to remove a portion thereof, and using the remaining photoresist material to protect the underlying portions of the substrate **210** during etching. In some embodiments, the fin **215** is formed on the substrate **210** by a growth process such as an epitaxial growth process.

[0025] In some embodiments, the dummy gate **270** includes polysilicon (also referred to as polycrystalline silicon or poly-Si or PO) or polycrystalline silicon-germanium (poly-SiGe). In some embodiments, the dummy gate **270** includes polycrystalline silicon. In some embodiments, the dummy gate **270** is formed by depositing a layer, e.g., a polycrystalline silicon layer, and patterning the deposited layer to form the dummy gate **270**. In some embodiments, the dummy gate is formed as one or more layers of one or more different materials.

[0026] In some embodiments, the gate spacers **240** are formed along sidewalls of the dummy gate **270** by depositing a dielectric layer on the substrate **210** to cover the dummy gate **270**, and then partially removing the deposited dielectric layer such that the gate spacers **240** remain along the sidewalls of the dummy gate **270**.

[0027] In some embodiments, the gate spacers **240** include a single layer structure. In some embodiments, the gate spacers **240** include a multilayer structure. In some embodiments, the gate spacers **240** include an insulating material. In some embodiments, the gate spacers **240** include silicon oxide, SiON, SiCN, SiOC, SiOCN, or SiN, or the like.

[0028] In some embodiments, forming the gate spacers **240** includes chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or the like.

[0029] In some embodiments, the partial removal of the deposited dielectric layer is performed using a process that includes an anisotropic etch operation. In some embodiments, the anisotropic etch operation forms the gate spacers **240** along the sidewalls of the dummy gate **270** by selectively removing horizontal portions of the dielectric layer relative to vertical portions of the dielectric layer, such that the etch operation results in vertically-oriented gate spacers **240** along the sidewalls of the dummy gate **270**. In some embodiments, the anisotropic etch operation results in upper portions of the gate spacers **240** having a rounded profile that curves towards the dummy gate **270**. In some embodiments, the forming of the gate spacers **240** along the sidewalls of the dummy gate **270** does not use a photoresist layer or pattern, or a lithography operation.

[0030] Referring to FIGS. **1** and **2B**, in an operation **120** portions of the fin **215** adjacent to the gate spacers **240** are removed to form recesses **215r** for source/drain regions.

[0031] In some embodiments, the gate spacers **240** are used to define a source/drain region (junction) profile. In some embodiments, the gate spacers **240** are used to offset doped regions from the gate structure for source/drain regions.

[0032] Referring to FIG. **2B**, portions of the fin **215** exposed by the dummy gate **270** and the gate spacers **240** are removed or recessed to form the recesses **215r** alongside the gate spacers **240**.

[0033] In some embodiments, removing the portions of the fin **215** includes forming a photoresist layer or a capping layer (such as an oxide capping layer) over the structure of FIG. **2A**, patterning the photoresist or capping layer to have openings that expose portions of the fin **215**, and using an etching process to etch the exposed portions of the fin **215**. In some embodiments, the fin **215** is etched using a dry etching process. In some embodiments, the etching process is a wet etching process, or combination dry and wet etching process. In some embodiments, removing the portions of the fin **215** includes a lithography process that includes coating a photoresist (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), other suitable processes, or the like. In some embodiments, removing the portions of the fin **215** includes maskless photolithography, electron-beam writing, ion-beam writing, or the like. In some embodiments, removing the portions of the fin **215** includes

a nanoimprint process. In some embodiments, a pre-cleaning process is performed to clean the recesses **215r** using an HF solution or the like.

[0034] Referring to FIGS. **1** and **2C**, in an operation **130**, after the portions of the fin **215** are removed alongside the gate spacers **240**, epitaxial layers **260** are formed in the recesses **215r** to form source/drain regions of the fin **215**, and an interlayer dielectric (ILD) layer **250** is formed on the substrate **210** at outer sides of the gate spacers **240**.

[0035] In some embodiments, the epitaxial layers **260** are formed using one or more epitaxy or epitaxial (epi) processes, such that Si features, SiGe features, or the like are formed in a crystalline state at the source/drain regions. In some embodiments, a lattice constant of the epitaxial layers **260** is different than a lattice constant of the fin **215** so that a channel region of the fin **215** is strained or stressed by the epitaxial layers **260**. In some embodiments, the strained or stressed channel region helps to improve carrier mobility and enhance the device performance. In some embodiments, the epitaxy process includes CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD)), molecular beam epitaxy, or the like. In some embodiments, the epitaxy process incorporates a dopant into the epitaxial layers **260**. In some embodiments, the epitaxy process uses gaseous or liquid precursors that interact with the material (e.g., silicon) of the fin **215** at the source/drain regions. In some embodiments, a strained channel region is implemented to increase carrier mobility and enhance device performance. In some embodiments, the epitaxial layers **260** are doped using P-type dopants such as boron or BF₃, N-type dopants such as phosphorus or arsenic, or the like. In some embodiments, a junction implant process is performed to dope the epitaxial layers **260**. In some embodiments, an annealing process, e.g., rapid thermal annealing (RTA) and/or laser annealing, is performed to activate the epitaxial layers **260**.

[0036] In some embodiments, the ILD layer **250** includes an insulating material such as silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, SiBN, SiCBN, tetraethoxysilane (TEOS), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), a low-k dielectric material, a combination thereof, or the like. Examples of low-k dielectric materials include fluorinated silica glass (FSG), carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), polyimide, and the like. In some embodiments, the ILD layer **250** includes a single layer. In some embodiments, the ILD layer **250** includes multiple layers.

[0037] In some embodiments, the ILD layer **250** is formed by CVD, ALD, spin-on-glass (SOG), or the like. In some embodiments, a chemical-mechanical planarization (CMP) process is used to planarize the ILD layer **250** and expose the top of the dummy gate **270**.

[0038] In FIG. **2C**, the dummy gate **270** and the gate spacers **240** are on the fin **215** and have a common bottom that is at about a same level as a top of the epitaxial layers **260**. The gate spacers **240** are along sidewalls of the dummy gate **270** and completely cover the sidewalls of the dummy gate **270**.

[0039] Referring to FIGS. **1** and **2D**, in an operation **140** a portion of the dummy gate **270** is removed, forming an opening **280** with the gate spacers **240** at sides thereof.

[0040] In some embodiments, the dummy gate **270** is partially removed using an etch process, e.g., a wet etch process or a dry etch process, that selectively removes the material of the dummy gate **270** relative to the material of the gate spacers **240**. In some embodiments, the wet etch process employs an etchant that includes ammonium hydroxide or the like. In some embodiments, the dummy gate **270** is partially removed using a dry etch process, or a combination of dry and wet etch processes. In some embodiments, the dummy gate **270** is partially removed using an anisotropic etch operation. In some embodiments, the material of the dummy gate **270** is partially removed using an etching process that includes a dry etch process using reaction gas(es) that selectively etch the dummy gate material relative to the materials of the gate spacers **240** and the ILD layer **250**. In some embodiments, the dummy gate **270** is partially removed without using a mask.

[0041] Referring to FIGS. **1** and **2E**, in an operation **150** a photoresist pattern 'PR' is formed on the

ILD layer **250** to serve as an etch mask. In FIG. 2E, the photoresist pattern PR has an opening that exposes upper surfaces of the partially removed dummy gate **270**, the gate spacers **240**, and a region of the ILD layer **250** beyond outer sidewalls **240so** of the gate spacers **240**.

[0042] In some embodiments, the photoresist pattern PR exposes the upper surfaces of the partially removed dummy gate **270** and the gate spacers **240** while lateral boundaries of the opening in the photoresist pattern PR are substantially aligned with the outer sidewalls **240so** of the gate spacers **240** (see also FIG. 4 and transistor structure II in FIG. 6B).

[0043] In some embodiments, the photoresist pattern PR exposes the upper surface of the dummy gate **270** while lateral boundaries of opening in the photoresist pattern PR are closer together than the outer sidewalls **240so** of the gate spacers **240** (see also FIG. 5 and transistor structure III in FIG. 6C).

[0044] Referring to FIGS. 1 and 2F, in an operation **160** the photoresist pattern PR is used as an etch mask in one or more etch processes that remove the remaining part of the dummy gate **270**, portions of the gate spacers **240**, and portions of the ILD layer **250**, resulting in a stepped opening or T-shaped trench **290**, and the photoresist pattern PR is removed.

[0045] In FIG. 2F, the T-shaped trench **290** has a wider upper portion **290u** and a narrower lower portion **290l**.

[0046] In some embodiments, a step where the upper portion **290u** of the T-shaped trench **290** transitions to the lower portion **290l** is sharp at corners **240c** of the gate spacers **240**. In some embodiments, the step is rounded at the corners **240c** due to etching.

[0047] In FIG. 2F, the removal of portions of the gate spacers **240** and portions of the ILD layer **250** results in the upper portion **290u** of the T-shaped trench **290** being bounded by sidewalls **250s** of the ILD layer **250**, with the sidewalls **250s** of the ILD layer **250** being spaced apart by a greater distance than the outer sidewalls **240so** of the gate spacers **240**, i.e., the ILD layer **250** is removed laterally beyond the outer sidewalls **240so** of the gate spacers **240** (see also transistor structure I in FIG. 6A). Thus, the sidewalls **250s** of the ILD layer **250** that form the sides of the upper portion **290u** of the T-shaped trench **290** are spaced apart by a distance that is greater than a distance between the outer sidewalls **240so** of the gate spacers **240**.

[0048] In some embodiments, the sidewalls **250s** of the ILD layer **250** are spaced apart by a same distance as the outer sidewalls **240so** of the gate spacers **240**, i.e., the ILD layer **250** is removed to be aligned with the outer sidewalls **240so** of the gate spacers **240** (see FIG. 4 and transistor structure II in FIG. 6B). That is, the sidewalls **250s** of the ILD layer **250** that form the side of the upper portion **290u** of the T-shaped trench **290** are substantially aligned with the outer sidewalls **240so** of the gate spacers **240**.

[0049] In some embodiments, the sidewalls **250s** of the ILD layer **250** are spaced apart by a distance that is less than the distance between the outer sidewalls **240so** of the gate spacers **240** and greater than a distance between inner sidewalls **240si** of the gate spacers **240**, while the gate spacers **240** remain for a substantial height of or a full height of the T-shaped trench **290** (see FIG. 5 and transistor structure III in FIG. 6C, in which the height of the gate spacers **240** is such that the gate spacers **240** extend vertically to vertically overlap an insulating layer such as a self-aligned contact (SAC) layer **235** (i.e., an imaginary horizontal line that passes through one of the gate spacers **240** also passes through the SAC layer **235**)). In this case, the step in the T-shaped trench **290** is formed in the gate spacers **240**, corresponding to a step from dimension W10 to dimension W20 of FIG. 5.

[0050] In FIG. 2F, the sidewalls **250s** of the ILD layer **250** are formed to be sloped at an angle D1 of greater than 90 degrees (with 90 degrees being vertical or parallel to the Z axis) using control of a sidewall removal process.

[0051] In some embodiments, the sidewalls **250s** of the ILD layer **250** are vertical, i.e., such that D1 is substantially 90 degrees (see, e.g., FIG. 4).

[0052] In some embodiments, the photoresist pattern PR is removed along with the remaining part

of the dummy gate **270**, portions of the gate spacers **240**, and portions of the ILD layer **250**. In some embodiments, the photoresist pattern PR is removed in a separate operation after removing the remaining part of the dummy gate **270**, portions of the gate spacers **240**, and portions of the ILD layer **250**.

[0053] In some embodiments, removing the remaining part of the dummy gate **270**, portions of the gate spacers **240**, and portions of the ILD layer **250** involves a first operation that removes the remaining part of the dummy gate **270** and a second operation that removes portions of the gate spacers **240** and portions of the ILD layer **250**. In some embodiments, the first operation precedes the second operation. In some embodiments, the first operation follows the second operation. In some embodiments, removing the remaining part of the dummy gate **270**, portions of the gate spacers **240**, and portions of the ILD layer **250** is performed using a wet etch or an isotropic etch. In some embodiments, the first and second operations are performed using different wet etch or isotropic etch chemistries.

[0054] Referring to FIGS. **1** and **2G**, in an operation **170** a replacement gate is formed. Operation **170** includes sequentially forming a gate dielectric layer **230**, a work function layer **225**, a filling conductor **220**, and the SAC layer **235**. In some embodiments, forming the gate dielectric layer **230**, the work function layer **225**, and the filling conductor **220** are operations that are included in a metal gate loop.

[0055] In FIG. **2G**, the gate dielectric layer **230** is formed on the fin **215**.

[0056] In some embodiments, the gate dielectric layer **230** is formed to line the entirety of the T-shaped trench **290** of FIG. **2F** that was formed by removing the remaining part of the dummy gate **270**, portions of the gate spacers **240**, and portions of the ILD layer **250**. In some embodiments, the gate dielectric layer **230** is formed to wrap an entire exposed area of the fin **215**. In some embodiments, the gate dielectric layer **230** is formed, e.g., patterned, to wrap a central portion of the fin **215** and expose a portion of the fin **215**.

[0057] In some embodiments, the gate dielectric layer **230** includes one or more layers of insulating material(s) such as silicon oxide or a high-k material. In some embodiments, the gate dielectric layer **230** includes a multilayer structure such as a layer of silicon oxide (e.g., as an interfacial layer) and another layer of a high-k material. In some embodiments, the gate dielectric layer **230** is a conformal layer. In some embodiments, a thickness of the gate dielectric layer **230** on the top of the fin **215** is different from a thickness of the gate dielectric layer **230** on a sidewall of the fin **215**.

[0058] In some embodiments, forming the gate dielectric layer **230** includes a process such as thermal oxidation, chemical vapor deposition, sputtering, or the like.

[0059] In some embodiments, the gate dielectric layer **230** includes one or more of a high-k dielectric material such as metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, or the like. In some embodiments, the gate dielectric layer **230** includes one or more of hafnium oxide (HfO₂), hafnium silicon oxide (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), lanthanum oxide (LaO), zirconium oxide (ZrO), titanium oxide (TiO), tantalum oxide (Ta₂O₅), yttrium oxide (Y₂O₃), strontium titanium oxide (SrTiO₃, STO), barium titanium oxide (BaTiO₃, BTO), barium zirconium oxide (BaZrO), hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (LaSiO), aluminum silicon oxide (AlSiO), aluminum oxide (Al₂O₃), silicon nitride (Si₃N₄), oxynitride (SiON), or the like. In some embodiments, the gate dielectric layer **230** is formed of a different material than the ILD layer **250**.

[0060] In some embodiments, forming the gate dielectric layer **230** is performed using chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), thermal oxide, ozone oxidation, combinations thereof, or the like.

[0061] In FIG. **2G**, the work function layer **225** is formed on the gate dielectric layer **230**.

[0062] In FIG. 2G, the lateral extent of the work function layer **225** extends beyond the outer sidewalls **240so** of the gate spacers **240**.

[0063] In some embodiments, the lateral extent of the work function layer **225** is narrower than the outer sidewalls **240so** of the gate spacers **240** but wider than the inner sidewalls **240si** of the gate spacers **240** (see, e.g., FIG. 5 and transistor structure III in FIG. 6C).

[0064] In some embodiments, the work function layer **225** adjusts a threshold voltage of the transistor. In some embodiments, the transistor is a P-type FET (PFET) and includes a P-type work function metal or metal-containing material such as TiN, TaC, TaN, Co, Ru, Mo, Al, or WN, or a silicide such as ZrSi.sub.2, MoSi.sub.2, TaSi.sub.2, or NiSi.sub.2, or other P-type work function layers, or combinations thereof. In some embodiments, the work function layer **225** includes a P-type work function material to provide a desired work function value for a P-type gate of a P-type semiconductor device. In some embodiments, the transistor is an N-type FET (NFET) and includes an N-type work function metal or metal-containing material such as Ti, Ag, Al, TiAl, TaAl, TaAlC, TaAlN, TaC, TaCN, TaSiN, Mn, or Zr, or other N-type work function layers, or combinations thereof. In some embodiments, the work function layer **225** includes an N-type work function material to provide a desired work function value for an N-type gate of an N-type semiconductor device.

[0065] In some embodiments, the work function layer **225** is a conformal layer that has a generally uniform thickness (width) throughout. In some embodiments, different portions of the work function layer **225** have different widths.

[0066] In some embodiments, the work function layer **225** is deposited using one or more of atomic layer deposition (ALD), evaporation, sputtering, chemical vapor deposition (CVD), PVD, remote plasma CVD (RPCVD), plasma enhanced CVD (PECVD), metal organic CVD (MOCVD), plating, combinations thereof, or the like. In some embodiments, the work function layer **225** is formed by conformal deposition of a work function material. In some embodiments, the work function material is deposited across an entire region of the substrate **210** and then removed from regions surrounding the gate structure using a CMP process, and then the work function material is partially removed from upper inner sidewalls of the gate spacers **240** by protecting a lower portion of the work function material, e.g., using a masking material, performing an etching process such as an etch back process, and then removing the masking material.

[0067] In FIG. 2G, the filling conductor **220** is formed on the work function layer **225**.

[0068] In FIG. 2G, the filling conductor **220** has a stepped or 'T' shape with a wider upper portion **220u** (cap) connected to a narrower lower portion **220l** (tail).

[0069] In some embodiments, the gate dielectric layer **230** and the work function layer **225** are conformal layers that conform to the sidewalls of the T-shaped trench **290** and have a substantially uniform thickness, and the filling conductor **220** is formed in the remaining volume of the T-shaped trench **290** such that the shape and slope of the sidewalls of the T-shaped trench **290** are reflected in the resultant shape and slope of the sidewalls of the filling conductor **220**.

[0070] In FIG. 2G, the filling conductor **220** overhangs the gate spacers **240**, i.e., a width, as determined parallel to the X axis, of the upper portion **220u** of the filling conductor **220** is greater than a distance, as determined parallel to the X axis, between the inner sidewalls **240si** of the gate spacers **240**.

[0071] In some embodiments, the filling conductor **220** is wider than in FIG. 2G and overhangs the ILD layer **250** above the gate spacers **240**, i.e., the lateral extent of the filling conductor **220** extends beyond the outer sidewalls **240so** of the gate spacers **240**.

[0072] In some embodiments, the filling conductor **220** is formed to a full height of the ILD layer **250**. In some embodiments, the filling conductor **220**, the work function layer **225**, and the gate dielectric layer **230** are partially removed using an etching process such as an etch back process, and the SAC layer **235** is formed in a space created by thereby.

[0073] In FIG. 2G, the lower portion **220l** of the filling conductor **220** is present in a lower portion

2251 of the work function layer **225**. The upper portion **220u** of the filling conductor **220** extends laterally across the work function layer **225**, which helps to increase a contact area between the filling conductor **220** and the work function layer **225** and helps to improve, i.e., reduce, gate resistance R_g .

[0074] In FIG. 2G, the filling conductor **220** and the work function layer **225** are laterally bounded by the gate dielectric layer **230**. In some embodiments, maintaining a lateral extent of the filling conductor **220** and the work function layer **225** within the boundaries of the gate dielectric layer **230** helps to reduce gate leakage and avoids shorts.

[0075] In some embodiments, filling conductor **220** is formed by filling in a region between inner sidewalls of the work function layer **225** such that the width of the filling conductor is a function of the width of the T-shaped trench **290**, the thickness of the gate dielectric layer **230**, and the thickness of the work function layer **225**.

[0076] In some embodiments, the filling conductor **220** includes one or more layers of conductive metal or metal-containing material(s). In some embodiments, the filling conductor **220** is or includes tungsten, aluminum, copper, or the like. In some embodiments, the filling conductor **220** is predominantly tungsten. In some embodiments, the material of the filling conductor **220** is more conductive than the material of the work function layer **225**.

[0077] In some embodiments, forming the filling conductor **220** includes a deposition process such as CVD, PVD, ALD, or the like, after which a CMP process is used to remove excess filling conductor **220** that overfills the T-shaped trench **290**.

[0078] In some embodiments, the top of the filling conductor **220** is processed to be recessed. In some embodiments, the top of the filling conductor **220** is recessed using an etchant that has high selectivity between the filling conductor **220** and the gate spacers **240**. In some embodiments, the top of the filling conductor **220** and the top of the work function layer **225** are processed to be recessed. In some embodiments, the top of the filling conductor **220**, the top of the work function layer **225**, and the top of the gate dielectric layer **230** are processed to be recessed. In some embodiments, a metal gate etch back (MGEb) process is used to recess the top of the gate structure and the gate dielectric layer **230**. In some embodiments, the metal gate etch back process includes a plasma etching process employing one or more etchants such as a fluorine-containing gas (e.g., one or more of CF_4 , SF_6 , CH_2F_2 , CHF_3 , or C_2F_6) and/or a chlorine-containing gas (e.g., one or more of Cl_2 , CHCl_3 , CCl_4 , BCl_3 , or SCl_4).

[0079] In FIG. 2G, the SAC layer **235** is formed on the recessed filling conductor **220**, work function layer **225**, and gate dielectric layer **230**.

[0080] In FIG. 2G, the SAC layer **235** is formed to be the full width of the filling conductor **220**, the work function layer **225**, and the gate dielectric layer **230**, i.e., the SAC layer **235** covers all of the filling conductor **220**, the work function layer **225**, and the gate dielectric layer **230**.

[0081] In FIG. 2G, the SAC layer **235** horizontally overlaps the gate spacers **240**. In other words, an imaginary vertical line that passes through one of the gate spacers **240** also passes through the SAC layer **235**.

[0082] In some embodiments, SAC layer **235** includes silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, a low-dielectric constant dielectric material, or the like. In some embodiments, SAC layer **235** includes silicon nitride.

[0083] In some embodiments, SAC layer **235** is formed using a deposition process, such as ALD, CVD, PVD, or the like. In some embodiments, after the deposition process, a CMP process is performed to remove portions of the SAC layer **235** that extend over the ILD layer **250**.

[0084] Referring to FIGS. 1 and 2H, in an operation **180** a resultant structure of operation **170** and FIG. 2G is processed to form source/drain (S/D) contacts that contact the epitaxial layers **260** adjacent to the gate.

[0085] FIG. 3A is an illustration of the transistor structure of FIG. 2G with annotations showing various dimensions and angles.

[0086] In FIG. 3A, a width of an upper portion of the gate structure (dimension W1) is greater than a width of a lower portion of the gate structure (dimension W2) (i.e., $W1 > W2$). In some embodiments, this structure allows for a greater width of the filling conductor **220** and helps to reduce gate resistance R_g while maintaining device performance. $W1 > W2$ provides for a relatively greater amount of the filling conductor **220** than a case where the entire gate structure has the width W2. In some embodiments, gate resistance R_g decreases with increased width W1. In some embodiments, increases in W1 are constrained by the need to avoid the gate structure becoming too close to an adjacent structure such as a contact. In some embodiments, the relationship of W1 to W2 decides a process window.

[0087] In FIG. 3A, upper portion of the gate structure overhangs the gate spacers **240** by at least a dimension DM-04. In some embodiments, DM-04 is sufficiently small to avoid the gate structure becoming too close to an adjacent structure, e.g., to avoid bridging to a source/drain contact.

[0088] In FIG. 3A, the fin **215** has sidewalls that are aligned with outer sidewalls of the gate spacers **240**.

[0089] In some embodiments, the gate spacers **240** have a lateral extent (dimension DM-40) that is less than a lateral extent of the sidewalls of the fin **215** such that the fin **215** is wider than dimension DM-40. In some embodiments, the gate spacers **240** have a lateral extent that is greater than that of the sidewalls of the fin **215** such that the fin **215** is narrower than the dimension DM-40 and the gate spacers **240** laterally overhang beyond the sidewalls of the fin **215**.

[0090] In FIG. 3A, an upper portion **225u** of the work function layer **225** has a lateral dimension (dimension DM-06) that is greater than the lateral extent of the gate spacers **240** (dimension DM-40).

[0091] In FIG. 3A, the upper portion **220u** of the filling conductor **220** has a lateral dimension (dimension DM-03) that is greater than a lateral extent of the lower portion **220l** of the filling conductor **220** (dimension DM-09). In other words, the upper portion **220u** of the filling conductor **220** is wider than the lower portion **220l** of the filling conductor. Here, “wider” refers to a dimension determined parallel to the X axis. Relative to a structure in which the filling conductor **220** is a single width throughout (e.g., having the width DM-09 throughout an entire height of the filling conductor **220**), the wider upper portion **220u** of the filling conductor **220** makes forming the lower portion **220l** easier by effectively reducing the aspect ratio of the filling conductor **220** as a whole, and helps lower gate resistance (R_g).

[0092] In some embodiments, the width of the lower portion **220l** of the filling conductor **220** (dimension DM-09) is substantially constant with increasing distance from the substrate.

[0093] In some embodiments, the upper portion **220u** of the filling conductor **220** is wider than the lower portion **220l** of the filling conductor for a substantial or entire length of the gate structure, i.e., a length in a Y axis direction in FIG. 3A (in/out of the page).

[0094] In some embodiments, a ratio of the width of the upper portion **220u** of the filling conductor **220** to the width of the lower portion **220l** of the filling conductor **220** (DM-03:DM-09) is substantially equal to a ratio of the width of the upper portion of the gate structure to the width of the lower portion of the gate structure ($W1:W2$).

[0095] In FIG. 3A, a sidewall angle of the upper portion **220u** of the filling conductor **220** is angle D5.

[0096] In some embodiments angles D1 and D5 are equal to one another, the width of the upper portion **220u** of the filling conductor is a constant fraction of the width W1 of the upper portion of the gate structure, and the filling conductor **220** has a cross sectional shape in the X-Z plane that corresponds to, while being smaller than, a cross-sectional shape of the gate structure in the X-Z plane. Enlarging the width of the upper portion of the gate structure and the upper portion **220u** of the filling conductor (i.e., enlarging W1 and DM-03) increases the amount and surface area of the filling conductor **220** and helps to reduce the gate resistance R_g .

[0097] In FIG. 3A, the upper portion **220u** of the filling conductor **220** has a lateral dimension

(dimension DM-03) that is greater than a distance between the inner sidewalls **240si** of the gate spacers **240** (see dimension W2). In other words, the upper portion **220u** of the filling conductor **220** is wider than the width W2 of the lower portion of the gate structure (i.e., wider than the distance between the inner sidewalls **240si** of the gate spacers **240**), and the upper portion **220u** of the filling conductor **220** overhangs the inner sidewalls **240si** of the gate spacers **240**.

[0098] In some embodiments, the upper portion **220u** of the filling conductor **220** has a lateral dimension (e.g., dimension DM-03 at the bottom or dimension DM-31 at the top of the upper portion **220u**) that is greater than a distance between the outer sidewalls **240so** of the gate spacers **240** (dimension DM-40).

[0099] In some embodiments, the upper portion **220u** of the filling conductor **220** has a lateral dimension (DM-03) that is greater than a lateral extent of the lower portion **220l** (dimension DM-09) but less than the distance between the inner sidewalls **240si** of the gate spacers **240** (see, e.g., FIGS. 4 and 5).

[0100] In FIG. 3A, the width of the upper portion **220u** of the filling conductor **220** transitions in a step manner to the width of the lower portion **220l** of the filling conductor **220**, i.e., there is a step from dimension DM-03 to dimension DM-09.

[0101] In some embodiments, the step in the filling conductor **220** is rounded. In some embodiments, the width of the upper portion **220u** of the filling conductor **220** transitions gradually or continuously to the width of the lower portion **220l** of the filling conductor **220**.

[0102] In FIG. 3A, the overall or outer width of the upper portion **225u** of the work function layer **225** transitions in a step manner to the overall or outer width of the lower portion **225l** of the work function layer **225**, i.e., there is a step from dimension DM-06 to dimension DM-50.

[0103] In some embodiments, the step in the work function layer **225** is rounded. In some embodiments, the overall or outer width of the upper portion **225u** of the work function layer **225** transitions in a curved step manner to the overall or outer width of the lower portion **225l** of the work function layer **225**, i.e., there is a curved step from dimension DM-06 to dimension DM-50, e.g., the upper inner corners **240c** of the gate spacers **240** are rounded, e.g., due to being rounded during an etching operation. In some embodiments, the overall or outer width of the upper portion **225u** of the work function layer **225** transitions gradually or continuously to the overall or outer width of the lower portion **225l** of the work function layer **225**.

[0104] In FIG. 3A, a height of the lower portion of the gate structure (dimension H2) and an overall height of the gate structure with the SAC layer **235** (see dimension H1) satisfy the relationship $H2/H1 \geq 1/3$. In some embodiments, maintaining $H2 \geq H1 * 1/3$ helps provide high device performance.

[0105] In FIG. 3A, the angle D1 of the sidewalls **250s** of the ILD layer **250** satisfies the relationship $120 \text{ degrees} \geq D1 \geq 90 \text{ degrees}$. The relationship $D1 \geq 90 \text{ degrees}$ allows the upper surface width of the upper portion **220u** of the filling conductor **220** (dimension DM-31) to be greater than the lower surface width of the upper portion **220u** of the filling conductor **220** (dimension DM-03) and simplifies formation of the filling conductor **220**. The relationship $D1 \leq 120 \text{ degrees}$ helps to maintain a distance between the gate structure and an adjacent structure such as a source/drain contact (see, e.g., FIG. 2H showing the source/drain contacts).

[0106] In FIG. 3A, the sidewall angle of the upper portion **220u** of the filling conductor **220** (the angle D5) satisfies the relationship $120 \text{ degrees} \geq D5 \geq 90 \text{ degrees}$. In some embodiments, $120 \text{ degrees} \geq D5 \geq 90 \text{ degrees}$ simplifies formation, e.g., filling, of the filling conductor **220**. In FIG. 3A, $D1 = D5$.

[0107] In FIG. 3A, $D1 > 90 \text{ degrees}$, which results in $DM-01 > DM-06$, i.e., the upper portion **225u** of the work function layer **225** gets wider (as determined parallel to the X axis) with increasing distance from the substrate. Also, $D5 > 90 \text{ degrees}$ results in $DM-31 > DM-03$, the upper portion **220u** of the filling conductor **220** gets wider (as determined parallel to the X axis) with increasing distance from the substrate.

[0108] In FIG. 3A, the gate spacers **240** have a height (DM-10) that corresponds to the height of the lower portion of the gate structure (H2).

[0109] FIG. 3B is an illustration of the transistor structure of FIG. 2G with annotations showing overlapping features.

[0110] In FIG. 3B, the upper portion **220u** of the filling conductor **220** is wider than a distance between the inner sidewalls **240si** of the gate spacers **240** such that the upper portion **220u** of the filling conductor **220** overhangs the gate spacers **240**. In other words, a first imaginary vertical line ln_a passes through the upper portion **220u** of the filling conductor **220** and one of the gate spacers **240** (it will be appreciated that the same is true for the opposite side of the gate structure in the case that the gate structure has mirror image symmetry). Herein, references to “vertical” refer to elements that are parallel to the Z axis.

[0111] In FIG. 3B, the work function layer **225** also overhangs the gate spacers **240**. In other words, a second imaginary vertical line ln_b passes through the upper portion **225u** of the work function layer **225** and one of the gate spacers **240** (it will be appreciated that the same is true for the opposite side of the gate structure in the case that the gate structure has mirror image symmetry).

[0112] In FIG. 3B, the work function layer **225** overhangs the ILD layer **250**. In other words, a third imaginary vertical line ln_c passes through the upper portion **225u** of the work function layer **225** and the ILD layer **250** (it will be appreciated that the same is true for the opposite side of the gate structure in the case that the gate structure has mirror image symmetry).

[0113] FIG. 4 is an illustration of a transistor structure according to some embodiments.

[0114] FIG. 4 is an illustration of a transistor structure according to some embodiments that generally corresponds to the transistor structure of FIGS. 2G, 3A, and 3B but has vertical sides aligned with the outer sidewalls **240so** of the gate spacers **240**. Elements not expressly described in connection with FIG. 4 are assumed to be substantially the same as those described above in connection with FIGS. 2G, 3A, and 3B.

[0115] In FIG. 4, an overall width $W10$ of the upper portion of the gate structure is substantially equal to an overall width of the lower portion of the gate structure including the gate spacers **240**. Thus, in FIG. 4, $W10=W2+2*W3$.

[0116] In FIG. 4, the sidewalls **250s** of the ILD layer **250** and the sides of the upper portion **220u** of the filling conductor **220** are vertical, i.e., having angles $D1$ and $D5$ that are substantially equal to 90 degrees. In FIG. 3A, $D1=D5$.

[0117] In some embodiments, $D1$ and $D5$ are equal to one another, and satisfy the relationships $120\text{ degrees} \geq D1 \geq 90\text{ degrees}$ and $120\text{ degrees} \geq D5 \geq 90\text{ degrees}$.

[0118] In FIG. 4, the upper portion **220u** of the filling conductor **220** has a width that is less than the distance between the inner sidewalls **240si** of the gate spacers **240**. In other words, an imaginary vertical line that passes through the upper portion **220u** of the filling conductor **220** would not pass through one of the gate spacers **240**.

[0119] In some embodiments, the upper portion **225u** of the work function layer **225** is made relatively thinner and the upper portion **220u** of the filling conductor **220** is made correspondingly larger, e.g., so that an imaginary vertical line does pass through the upper portion **220u** of the filling conductor **220** and one of the gate spacers **240** (see imaginary line ln_a in FIG. 3B).

[0120] In FIG. 4, the upper portion **220u** of the filling conductor **220** is wider than the lower portion **220l** of the filling conductor, which makes forming the lower portion **220l** easier and helps to reduce gate resistance R_g while maintaining device performance.

[0121] In FIG. 4, the gate spacers **240** have a height (dimension DM-10) that corresponds to the height of the lower portion of the gate structure, i.e., the gate spacers **240** are not a full height of the gate structure. In some embodiments, the height of the gate spacers **240** being less than a full height of the gate structure simplifies fabrication of the gate by allowing for greater variation in etch characteristics, due to the gate spacers **240** having a greater thickness overall.

[0122] FIG. 5 is an illustration of a transistor structure according to some embodiments.

[0123] FIG. 5 is an illustration of a transistor structure in which outer sidewalls of the upper portion of the gate structure are closer together than the outer sidewalls **240so** of the gate spacers **240** but farther apart than the inner sidewalls **240si** of the gate spacers **240**. Elements not expressly described in connection with FIG. 5 are assumed to be substantially the same as those described above in connection with FIGS. 3B and 4.

[0124] In FIG. 5, the sides of the upper portion **220u** of the filling conductor **220** are vertical, i.e., the angle D5 is substantially equal to 90 degrees. Also, the gate spacers **240** have a step therein that is substantially square such that an enclosed angle D10 of the step is substantially equal to 90 degrees. In FIG. 5, D10=D5.

[0125] In some embodiments, D10 and D5 are equal to one another and satisfy the relationships $120\text{ degrees} \geq D10 \geq 90\text{ degrees}$ and $120\text{ degrees} \geq D5 \geq 90\text{ degrees}$.

[0126] In FIG. 5, the upper portion **220u** of the filling conductor **220** has a width that is less than the distance between the inner sidewalls **240si** of the gate spacers **240**. In other words, an imaginary vertical line that passes through the upper portion **220u** of the filling conductor **220** would not pass through one of the gate spacers **240**.

[0127] In some embodiments, the upper portion **225u** of the work function layer **225** is made relatively thinner and the upper portion **220u** of the filling conductor **220** is made correspondingly larger, e.g., so that an imaginary vertical line does pass through the upper portion **220u** of the filling conductor **220** and one of the gate spacers **240** (see imaginary line ln_a in FIG. 3B).

[0128] In FIG. 5, the upper portion **220u** of the filling conductor **220** is wider than the lower portion **220l** of the filling conductor, which makes forming the lower portion **220l** easier and helps to reduce gate resistance R_g while maintaining device performance.

[0129] In FIG. 5, the gate spacers **240** have a height that is sufficient to cover substantially the full height of the gate structure. Thus, the entirety of the gate dielectric layer **230** is separated from the surrounding ILD layer **250** by the gate spacers **240**. In some embodiments, the entirety of the gate dielectric layer **230** being separated from the surrounding ILD layer **250** by the gate spacers **240** helps to improve isolation of the gate.

[0130] FIGS. 6A-6C are cross-sectional views of transistor structures according to some embodiments.

[0131] FIGS. 6A-6C include three photoresist patterns PR and three corresponding transistor structures.

[0132] The photoresist pattern PR of FIG. 6A corresponds to the previously-described photoresist pattern PR in FIG. 2E.

[0133] The photoresist pattern PR in FIG. 6A has an opening with lateral boundaries that are spaced apart by a distance dI (as determined parallel to the X axis). The photoresist pattern PR in FIG. 6B has an opening with lateral boundaries that are spaced apart by a distance dII , which is less than the distance dI . The lateral boundaries of the opening in the photoresist pattern PR in FIG. 6B are substantially aligned with the outer sidewalls of the gate spacers **240**. The photoresist pattern PR in FIG. 6C has an opening with lateral boundaries that are spaced apart by a distance $dIII$, which is less than the distance dII .

[0134] The photoresist pattern PR in FIG. 6A is used to form the transistor structure I in which the upper portion of the gate structure is wider than the outer sidewalls **240so** of the gate spacers **240**. The photoresist pattern PR in FIG. 6B is used to form the transistor structure II in which the upper portion of the gate structure has a same width as the outer sidewalls **240so** of the gate spacers **240**. The photoresist pattern PR in FIG. 6C is used to form the transistor structure III in which the upper portion of the gate structure is narrower than the outer sidewalls **240so** of the gate spacers **240** but wider than the inner sidewalls **240si** of the gate spacers **240**.

[0135] In some embodiments of the transistor structure I, the upper portion **225u** of the work function layer **225** is wider than the outer sidewalls **240so** of the gate spacers **240**. In some

embodiments of the transistor structure I, the upper portion **220u** of the filling conductor **220** is wider than the outer sidewalls **240so** of the gate spacers **240**.

[0136] In some embodiments of the transistor structure II, the upper portion **225u** of the work function layer **225** is wider than the inner sidewalls **240si** of the gate spacers **240**. In some embodiments of the transistor structure II, the upper portion **220u** of the filling conductor **220** is wider than the inner sidewalls **240si** of the gate spacers **240**.

[0137] In some embodiments of the transistor structure III, the upper portion **225u** of the work function layer **225** is wider than the inner sidewalls **240si** of the gate spacers **240**. In some embodiments of the transistor structure III, the upper portion **220u** of the filling conductor **220** is wider than the inner sidewalls **240si** of the gate spacers **240**.

[0138] In transistor structures I and II, $120 \text{ degrees} \geq D1 \geq 90 \text{ degrees}$, and in transistor structure III, $120 \text{ degrees} \geq D10 \geq 90 \text{ degrees}$. Maintaining the angles D1 and D10 at 120 degrees or less helps to avoid issues with adjacent structures, e.g., helps to avoid bridging with an adjacent contact such as a source/drain contact. Maintaining the angles D1 and D10 at 90 degrees or more helps with filling the filling conductor **220**.

[0139] In transistor structures I, II, and III, the SAC layer **235** overlaps the gate spacers **240**. In other words, an imaginary vertical line that passes through one of the gate spacers **240** also passes through the SAC layer **235**.

[0140] In some embodiments, a method of fabricating a semiconductor device includes forming a dummy gate and gate spacers on a semiconductor region of a substrate; forming an interlayer dielectric layer along sides of the gate spacers; forming a trench having a lower portion and an upper portion, the upper portion of the trench being wider than the lower portion of the trench, forming the trench including: removing a part of the gate spacers, and removing the dummy gate; and forming a gate structure in the trench. The gate structure has a lower portion and an upper portion, the upper portion of the gate structure being in the upper portion of the trench, the lower portion of the gate structure being in the lower portion of the trench, and the upper portion of the gate structure being wider than the lower portion of the gate structure, a first one of the gate spacers has a first inner sidewall that faces a first side of the lower portion of the gate structure, and a second one of the gate spacers has a second inner sidewall that faces a second side of the lower portion of the gate structure, the first inner sidewall being spaced apart from the second inner sidewall by a first distance, and the upper portion of the gate structure has a width that is greater than the first distance. In some embodiments, forming the trench includes removing a part of the interlayer dielectric layer, sides of the upper portion of the trench are defined by sidewalls of the interlayer dielectric layer, and a minimum separation of the sidewalls is greater than a distance between outer sides of the gate spacers. In some embodiments, forming the trench includes performing at least one etch operation to remove the part of the interlayer dielectric layer, the part of the gate spacers, and the dummy gate, and the part of the interlayer dielectric layer is etched so that the upper portion of the trench is wider than the distance between outer sides of the gate spacers. In some embodiments, forming the trench includes forming the sidewalls to have an angle D1 relative to a major plane of a surface of the substrate that ranges from 90 degrees to 120 degrees. In some embodiments, the upper portion of the gate structure is formed to have sides that have the angle D1 relative to the major plane of the surface of the substrate. In some embodiments, D1 is greater than 90 degrees. In some embodiments, the method further includes forming an insulating layer on top of the gate structure, and the gate spacers have a height that is at least $\frac{1}{3}$ of an overall distance from a bottom of the gate spacers to a top of the insulating layer. In some embodiments, forming the trench includes removing a part of the interlayer dielectric layer, sides of the upper portion of the trench are defined by sidewalls of the interlayer dielectric layer, and a minimum separation of the sidewalls is approximately the same as a distance between outer sides of the gate spacers. In some embodiments, forming the trench includes performing at least one etch operation to remove the part of the interlayer dielectric layer, the part of the gate spacers, and the

dummy gate, and the part of the interlayer dielectric layer is etched so that the sidewalls of the interlayer dielectric layer in the upper portion of the trench are substantially aligned with outer sides of the gate spacers. In some embodiments, the method further includes forming an insulating layer on top of the gate structure, and the gate spacers have a height that is at least $\frac{1}{3}$ of an overall distance from a bottom of the gate spacers to a top of the insulating layer. In some embodiments, forming the trench includes making upper portions of gate spacers narrower, and sides of the upper portion of the trench are defined by sidewalls of upper portions of the gate spacers. In some embodiments, forming the trench includes performing at least one etch operation to remove the part of the gate spacers and the dummy gate, and the gate spacers are etched so that the sidewalls of the upper portion of the trench are aligned with a point between inner sides and outer sides of the gate spacers. In some embodiments, the gate structure includes a filling conductor, the filling conductor is formed to have a lower portion and an upper portion, the upper portion of the filling conductor is in the upper portion of the trench, the lower portion of the filling conductor is in the lower portion of the trench, and the upper portion of the filling conductor is wider than the lower portion of the filling conductor. In some embodiments, forming the gate structure includes: forming a gate dielectric layer, the gate dielectric layer being formed to have a substantially uniform first thickness; forming a work function layer on the gate dielectric layer, the work function layer being formed to have a substantially uniform second thickness; and forming a filling conductor on the work function layer, the filling conductor being formed to fill a remaining part of the trench not occupied by the gate dielectric layer and the work function layer, and having a greater electrical conductivity than the work function layer.

[0141] In some embodiments, a semiconductor device includes: a semiconductor region having a channel region therein; a gate structure adjacent to the channel region; a first gate spacer at a first side of the gate structure and a second gate spacer at a second side of the gate structure; and an interlayer dielectric layer along sides of the first and second gate spacers. The gate structure has a lower portion and an upper portion, the upper portion of the gate structure being wider than the lower portion of the gate structure, the first gate spacer has a first inner sidewall that faces a first side of the lower portion of the gate structure, and the second gate spacer has a second inner sidewall that faces a second side of the lower portion of the gate structure, the first inner sidewall being spaced apart from the second inner sidewall by a first distance, and the upper portion of the gate structure has a width that is greater than the first distance. In some embodiments, the upper portion of the gate structure has a width that is greater than a distance between outer sidewalls of the first and second gate spacers. In some embodiments, the upper portion of the gate structure has a width that is approximately the same as a distance between outer sides of the first and second gate spacers. In some embodiments, the first and second gate spacers each have a wider bottom portion and a narrower top portion with a substantially horizontal step between the bottom portion and the top portion, and outer sidewalls of the upper portion of the gate structure contact inner sidewalls of the top portions of the first and second gate spacers. In some embodiments, the semiconductor device further includes an insulating layer on top of the gate structure, and the first and second gate spacers have a height that is at least $\frac{1}{3}$ of an overall distance from a bottom of the first and second gate spacers to a top of the insulating layer.

[0142] In some embodiments, a semiconductor device includes a semiconductor region; a gate structure; an insulating layer of a dielectric material contacting a top of the gate structure and extending across a full width of the top of the gate structure; a first gate spacer and a second gate spacer; and an interlayer dielectric layer along sides of the first and second gate spacers. The gate structure has a lower portion and an upper portion, the lower portion extending vertically between the first and second gate spacers and having a first width, and the upper portion extending vertically from the lower portion to the insulating layer, and having a second width that is greater than the first width, the first gate spacer has a first inner sidewall that faces a first side of the lower portion of the gate structure, and the second gate spacer has a second inner sidewall that faces a

second side of the lower portion of the gate structure, the first inner sidewall being spaced apart from the second inner sidewall by a first distance, and the insulating layer has a width that is greater than the first distance.

[0143] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of fabricating a semiconductor device, the method comprising: forming a dummy gate and gate spacers on a semiconductor region of a substrate; forming an interlayer dielectric layer along sides of the gate spacers; forming a trench having a lower portion and an upper portion, the upper portion of the trench being wider than the lower portion of the trench, forming the trench including: removing a part of the gate spacers, and removing the dummy gate; and forming a gate structure in the trench, wherein: the gate structure has a lower portion and an upper portion, the upper portion of the gate structure being in the upper portion of the trench, the lower portion of the gate structure being in the lower portion of the trench, and the upper portion of the gate structure being wider than the lower portion of the gate structure, a first one of the gate spacers has a first inner sidewall that faces a first side of the lower portion of the gate structure, and a second one of the gate spacers has a second inner sidewall that faces a second side of the lower portion of the gate structure, the first inner sidewall being spaced apart from the second inner sidewall by a first distance, and the upper portion of the gate structure has a width that is greater than the first distance.
2. The method of claim 1, wherein: forming the trench includes removing a part of the interlayer dielectric layer, sides of the upper portion of the trench are defined by sidewalls of the interlayer dielectric layer, and a minimum separation of the sidewalls is greater than a distance between outer sides of the gate spacers.
3. The method of claim 2, wherein: forming the trench includes performing at least one etch operation to remove the part of the interlayer dielectric layer, the part of the gate spacers, and the dummy gate, and the part of the interlayer dielectric layer is etched so that the upper portion of the trench is wider than the distance between outer sides of the gate spacers.
4. The method of claim 2, wherein forming the trench includes forming the sidewalls to have an angle D1 relative to a major plane of a surface of the substrate that ranges from 90 degrees to 120 degrees.
5. The method of claim 4, wherein the upper portion of the gate structure is formed to have sides that have the angle D1 relative to the major plane of the surface of the substrate.
6. The method of claim 5, wherein D1 is greater than 90 degrees.
7. The method of claim 2, further comprising forming an insulating layer on top of the gate structure, wherein the gate spacers have a height that is at least $\frac{1}{3}$ of an overall distance from a bottom of the gate spacers to a top of the insulating layer.
8. The method of claim 1, wherein: forming the trench includes removing a part of the interlayer dielectric layer, sides of the upper portion of the trench are defined by sidewalls of the interlayer dielectric layer, and a minimum separation of the sidewalls is approximately the same as a distance between outer sides of the gate spacers.
9. The method of claim 8, wherein: forming the trench includes performing at least one etch

operation to remove the part of the interlayer dielectric layer, the part of the gate spacers, and the dummy gate, and the part of the interlayer dielectric layer is etched so that the sidewalls of the interlayer dielectric layer in the upper portion of the trench are substantially aligned with outer sides of the gate spacers.

10. The method of claim 8, further comprising forming an insulating layer on top of the gate structure, wherein the gate spacers have a height that is at least $\frac{1}{3}$ of an overall distance from a bottom of the gate spacers to a top of the insulating layer.

11. The method of claim 1, wherein: forming the trench includes making upper portions of gate spacers narrower, and sides of the upper portion of the trench are defined by sidewalls of upper portions of the gate spacers.

12. The method of claim 11, wherein: forming the trench includes performing at least one etch operation to remove the part of the gate spacers and the dummy gate, and the gate spacers are etched so that the sidewalls of the upper portion of the trench are aligned with a point between inner sides and outer sides of the gate spacers.

13. The method of claim 1, wherein: the gate structure includes a filling conductor, the filling conductor is formed to have a lower portion and an upper portion, the upper portion of the filling conductor is in the upper portion of the trench, the lower portion of the filling conductor is in the lower portion of the trench, and the upper portion of the filling conductor is wider than the lower portion of the filling conductor.

14. The method of claim 1, wherein forming the gate structure includes: forming a gate dielectric layer, the gate dielectric layer being formed to have a substantially uniform first thickness; forming a work function layer on the gate dielectric layer, the work function layer being formed to have a substantially uniform second thickness; and forming a filling conductor on the work function layer, the filling conductor being formed to fill a remaining part of the trench not occupied by the gate dielectric layer and the work function layer, and having a greater electrical conductivity than the work function layer.

15. A semiconductor device comprising: a semiconductor region having a channel region therein; a gate structure adjacent to the channel region; a first gate spacer at a first side of the gate structure and a second gate spacer at a second side of the gate structure; and an interlayer dielectric layer along sides of the first and second gate spacers, wherein: the gate structure has a lower portion and an upper portion, a width of the upper portion of the gate structure being different from a width of the lower portion of the gate structure, the first gate spacer has a first inner sidewall that faces a first side of the lower portion of the gate structure, and the second gate spacer has a second inner sidewall that faces a second side of the lower portion of the gate structure, the first inner sidewall being spaced apart from the second inner sidewall by a first distance, and the upper portion of the gate structure has a width that is greater than the first distance.

16. The semiconductor device of claim 15, wherein the upper portion of the gate structure has a width that is greater than a distance between outer sidewalls of the first and second gate spacers.

17. The semiconductor device of claim 15, wherein the upper portion of the gate structure has a width that is approximately the same as a distance between outer sides of the first and second gate spacers.

18. The semiconductor device of claim 15, wherein: the first and second gate spacers each have a wider bottom portion and a narrower top portion with a substantially horizontal step between the bottom portion and the top portion, and outer sidewalls of the upper portion of the gate structure contact inner sidewalls of the top portions of the first and second gate spacers.

19. The semiconductor device of claim 15, further comprising an insulating layer on top of the gate structure, wherein the first and second gate spacers have a height that is at least $\frac{1}{3}$ of an overall distance from a bottom of the first and second gate spacers to a top of the insulating layer.

20. A semiconductor device comprising: a semiconductor region; a gate structure on the semiconductor region; an insulating layer of a dielectric material contacting a top of the gate

structure and extending across a full width of the top of the gate structure; a first gate spacer and a second gate spacer; and an interlayer dielectric layer along sides of the first and second gate spacers, wherein: the gate structure has a lower portion and an upper portion, the lower portion extending vertically between the first and second gate spacers and having a first width, and the upper portion extending vertically from the lower portion to the insulating layer, and having a second width that is greater than the first width, the first gate spacer has a first inner sidewall that faces a first side of the lower portion of the gate structure, and the second gate spacer has a second inner sidewall that faces a second side of the lower portion of the gate structure, the first inner sidewall being spaced apart from the second inner sidewall by a first distance, and the insulating layer has a width that is greater than the first distance.
